that was overlapped by Explorer. If you were to move the Explorer window to show the entire browser, Windows would make an API call asking for the entire frame to be redrawn. This lack of information also made transparencies a difficult task to execute (there are workarounds to this, however).

Clearly, innovations were needed both on the hardware and the software level to better present interface information. These innovations first came about as Windows "accelerators." These were hardware cards that would offload the most basic of rendering tasks from the CPU—tasks such as drawing lines and circles. These accelerators slowly gave way to graphic cards with more complex transformation and rendering strengths (NVIDIA's GeForce series introduced such functionalities to consumer hardware). Today, a graphics card is powered by a processor powerful enough to handle complex mathematical calculations.

These cards also come with enough memory to dwarf hard drive capacities of old. Of course, the old technology of GDI and the GDI+ bandage that MS applied to the technology is woefully inadequate to leverage the graphical power at the operating system's disposal. And this state persists with Windows XP: resolution-independent interfaces are out of the question, and applications have to run through multiple hoops and myriad "solutions" to offer visual effects such as window animations, transparencies, and window scaling.

2.2.3 Accelerating Windows

To address several of these shortcomings, Vista offers a completely new way to talk to the video hardware and draw onscreen elements. The biggest change is that it has done away with a raster-based (bit-mapped) API—Vista offers APIs which make use of vector-based drawings to draw elements of the window. A vector is essentially a mathematical formula used to define an object. So while a raster image of a 20-pixel-wide circle would comprise of a fixed number of pixels drawn on screen, a vector image of the same circle would just tell the computer to draw a circle with a diameter